

CSE 465

Mobile Computing

1. Mobile Computing Introduction

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What is Mobile Computing?

- "A computing paradigm that enables **communication, computation, and sensing** for users and devices that are **physically mobile**."
- Aim to support continuous operation under
 - Changing network conditions (WiFi, LTE, 5G)
 - Limited energy and hardware resources
 - Dynamic user context and environment
- Key characteristics
 - Wireless connectivity
 - Battery-powered devices
 - Resource constraints (CPU, memory, energy)
 - Context-aware sensing
 - Seamless cloud-edge-device interaction



Mobile vs. Ubiquitous vs. Pervasive Computing



Mobile Computing

CORE CONCEPT

"The computer moves with the user."

KEY FOCUS

Portability, wireless connectivity, and overcoming device resource constraints.

USER INTERACTION

Explicit: The user actively operates a distinct, portable device.

PRIMARY EXAMPLE

Smartphones, Laptops, Tablets, Wearables.



Ubiquitous Computing

CORE CONCEPT

"The computer disappears into the background."

KEY FOCUS

Invisibility, calm technology, and seamless integration into the environment.

USER INTERACTION

Implicit: The environment senses the user and reacts automatically.

PRIMARY EXAMPLE

Smart rooms, automated climate/lighting based on human presence.



Pervasive Computing

CORE CONCEPT

"Computing is embedded in everyday objects."

KEY FOCUS

Interconnectedness, microchips everywhere, and continuous data exchange (IoT).

USER INTERACTION

Distributed: Devices constantly communicate with each other and users.

PRIMARY EXAMPLE

Smart appliances, RFID supply chains, connected vehicles.

Mobile vs. Ubiquitous vs. Pervasive Computing



Mobile
Computing

CORE CONCEPT



Ubiquitous
Computing

CORE CONCEPT



Pervasive
Computing

CORE CONCEPT

Modern Perspective (for Mobile AI):
Today, their **boundaries blur!**

- Mobile computing → **infrastructure**
- Ubiquitous computing → **vision**
- Pervasive computing → **behavior & intelligence**

Explicit: The user actively operates a distinct, portable device.

PRIMARY EXAMPLE

Smartphones, Laptops, Tablets, Wearables.

Implicit: The environment senses the user and reacts automatically.

PRIMARY EXAMPLE

Smart rooms, automated climate/lighting based on human presence.

Distributed: Devices constantly communicate with each other and users.

PRIMARY EXAMPLE

Smart appliances, RFID supply chains, connected vehicles.

Why Mobile Computing Matters?

- **Computing Happens Where Humans Are**

- Smartphones, wearables, vehicles, robots
- Always with the user
- Enables real-time interaction

- **Massive Global Scale**

- ~7 Billions of smartphones worldwide (2025)
- ~21 Billions of IoT devices (2025)
- Dominant computing platform globally!

- **Foundation of Future Intelligent Systems**

- Robots and autonomous systems face the same constraints: limited compute, battery, wireless connectivity
- Intelligence must run at the edge with real-time perception and control
- Large-scale deployment requires distributed, energy-efficient design



Mobile Computing Characteristics

- **Mobility**
 - Devices and users change physical location
 - Dynamic network attachment and handoff
- **Wireless Communication**
 - Variable bandwidth and latency
 - Intermittent connectivity
- **Resource Constraints**
 - Limited battery capacity
 - Limited CPU, GPU/NPU, and memory
- **Context Awareness**
 - Location, activity, environment sensing
 - Adaptive system behavior
- **Distributed Architecture**
 - Device ↔ Edge ↔ Cloud collaboration



Mobile Computing Challenges

- **Mobility & Disconnection**

- Frequent handoffs
- Intermittent connectivity
- Variable network quality

- **Resource Constraints**

- Limited battery
- Limited CPU, memory, storage
- Wireless bandwidth is scarce and variable

- **Scalability**

- Massive numbers of users and devices
- Need for decentralized system design

- **Security & Privacy**

- Sensitive personal data
- Increased attack surface in wireless systems

- **Adaptation**

- Systems must dynamically adapt to:
 - Network conditions
 - User context
 - Resource availability

- **Reference**

- Satyanarayanan, Mahadev. "Pervasive computing: Vision and challenges." IEEE Personal Communications 8.4 (2001): 10-17.



Mobile Computing Challenges

Evolution of Mobile Computing

1991

Ubiquitous Computing

- Invisible technology
- Human-centered design
- Desktop to environment shift



Mid-1990s

Mobility & Distributed Systems

- Weak connectivity issues
- Disconnected operation
- Mobility support



Early 2000s

Mobile Information Access

- Bandwidth variability
- Resource adaptation
- Mobile-aware apps



2007-2012

Smartphone Revolution

- iPhone & Android OS
- App-centric computing
- Always-on connectivity

Late '00s - '10s

Cloud Offloading & Edge

- Resource augmentation
- Computation offloading
- Cloudlets utilization



2010s

Participatory Sensing

- Smartphones as sensor hubs
- Human activity recognition
- Behavioral data collection



Late 2010s

Deep Learning on Mobile

- Model compression techniques
- NPU hardware acceleration
- On-device vision/audio



2020s (Present)

On-Device AI & Mobile LLMs

- Direct on-device inference
- Edge-cloud collaboration
- Personalized AI agents

Evolution of Mobile Computing

- **Early Vision: Ubiquitous Computing (1991)**

- Computing embedded into everyday life
- Calm, invisible technology
- Human-centered design

- Key idea: The most profound technologies disappear into the background.

- Impact:

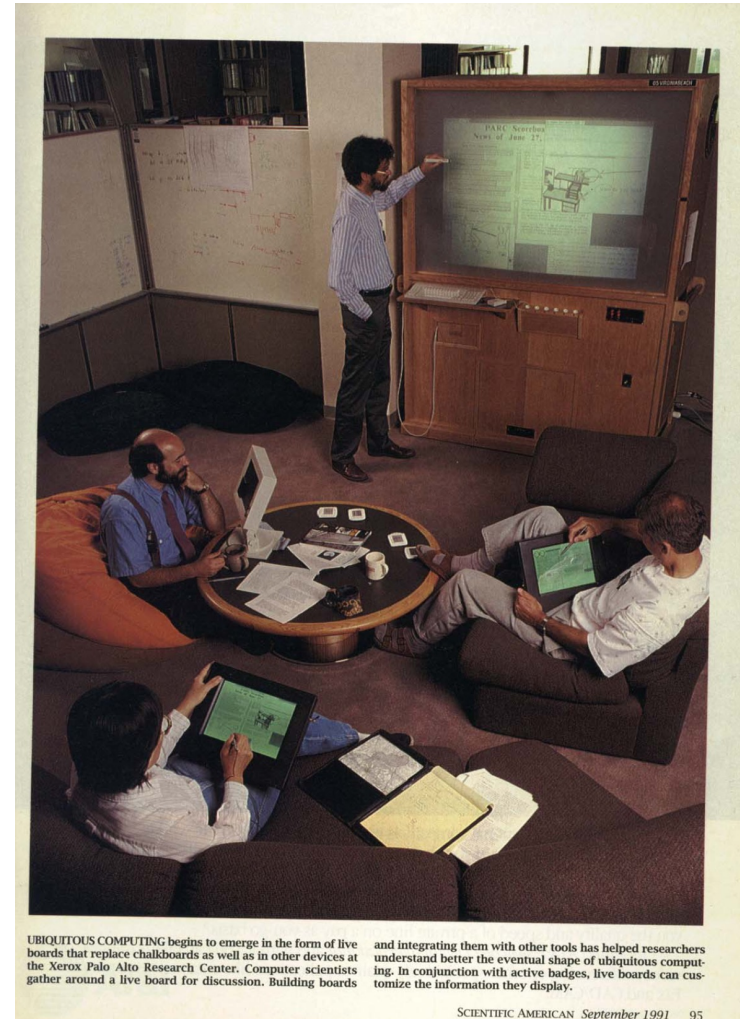
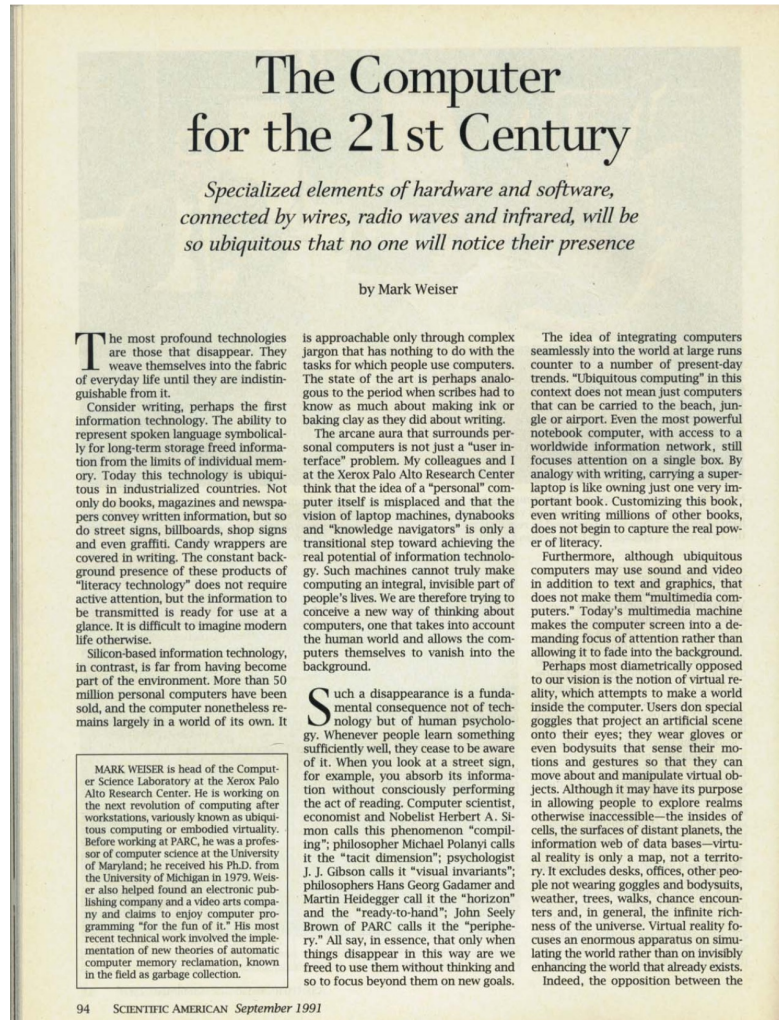
- Shift from desktop → environment-integrated computing
- Foundation of mobile and pervasive systems research

- **Reference**

Weiser, Mark. "The computer for the 21st century." Scientific American (1991)

Evolution of Mobile Computing

- Weiser, Mark. "The computer for the 21st century." Scientific American (1991)



Evolution of Mobile Computing

- **Mobility & Distributed Systems (Mid-1990s)**

- Core challenge:

- How to support computing under mobility and disconnection?

- Key concepts:

- Weak connectivity
- Disconnected operation
- Mobile file systems

- Representative work:

- Coda distributed file system
- Optimistic replication

- **Reference**

Satyanarayanan, Mahadev, et al. "Coda: A highly available file system for a distributed workstation environment." IEEE Transactions on Computers 39.4 (1990): 447-459.



Evolution of Mobile Computing

- **Mobile Information Access (Early 2000s)**

- Shift:

- From file systems → mobile-aware applications

- Key problems:

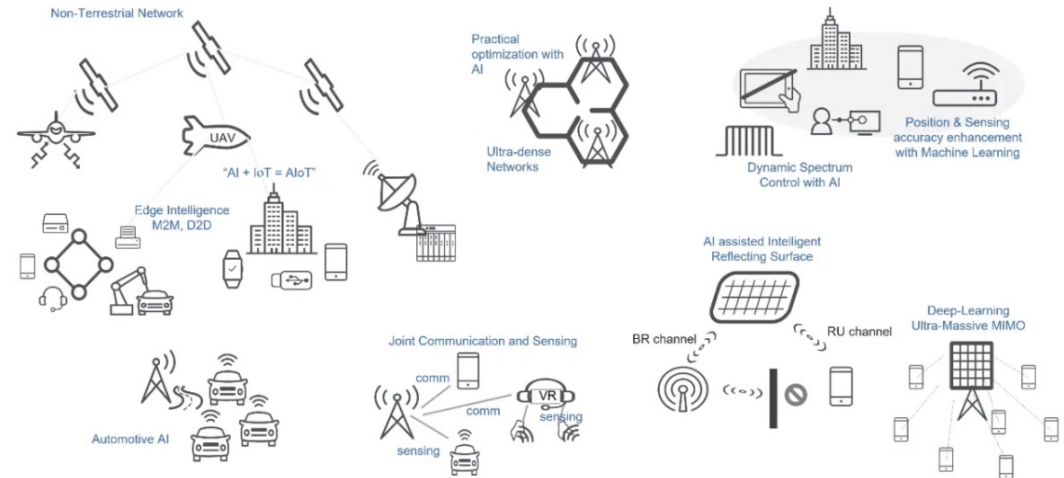
- Bandwidth variability
- Latency sensitivity
- Resource adaptation

- Vision:

- Support seamless information access while moving

- **Reference**

Satyanarayanan, Mahadev. "Pervasive computing: Vision and challenges." IEEE Personal Communications 8.4 (2001): 10-17.



Evolution of Mobile Computing

- **Smartphone Revolution (2007-2012)**

- Turning point:

- iPhone (2007)
- Android ecosystem

- Impact:

- App-centric computing
- Integrated multi-sensor platforms
- Massive developer ecosystem

- Research focus:

- Mobile app performance
- Energy optimization
- Mobile sensing



Evolution of Mobile Computing

- **Cloud Offloading & Edge Computing (Late 2000s - Early 2010s)**

- Problem:
 - Mobile devices are resource constrained
- Solution:
 - Offload computation to cloud
- Key idea:
 - Augment mobile devices with remote servers
- Representative work:
 - Cloudlets

- **Reference**

Satyanarayanan, Mahadev, et al. "The case for VM-based cloudlets in mobile computing." IEEE Pervasive Computing 8.4 (2009): 14-23.



Evolution of Mobile Computing

- **Mobile Sensing & Participatory Sensing (2010s)**

- Smartphones as sensor hubs:

- Accelerometer
- Gyroscope
- GPS
- Camera
- Microphone

- **New paradigm:**

- Large-scale behavioral data collection
- Human activity recognition

- **Reference**

- Lane, Nicholas D., et al. "A survey of mobile phone sensing." IEEE Communications Magazine 48.9 (2010): 140-150.
- E. Miluzzo, et al., "Darwin Phones: the Evolution of Sensing and Inference on Mobile Phones," ACM MobiSys 2010.



Evolution of Mobile Computing

- **Deep Learning on Mobile (Late 2010s)**

- New challenge:

- Neural networks are computation-heavy

- Research directions:

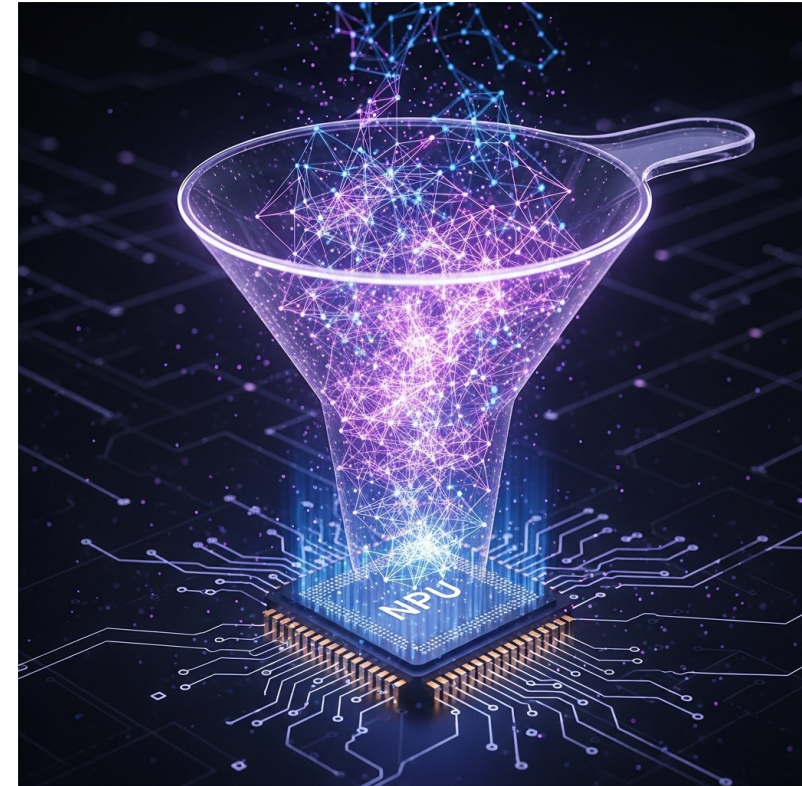
- Model compression
- Quantization
- Hardware acceleration (NPU)
- Efficient inference

- Mobile computing becomes:

→ AI-centric systems research

- **Reference**

Sze, Vivienne, et al. "Efficient processing of deep neural networks: A tutorial and survey." Proceedings of the IEEE 105.12 (2017): 2295-2329.



Evolution of Mobile Computing

- **On-Device AI & Mobile LLMs (2020s)**
- Current frontier:
 - On-device inference
 - Personalized AI
 - Edge-cloud collaboration
 - Mobile agents
- Trends:
 - LLM compression
 - Edge acceleration
 - Continual learning
- Mobile computing today:
 - Infrastructure for scalable, personalized AI systems



Why This Course Focuses on Mobile AI

- **Mobile computing has evolved toward intelligence at the edge**
 - Today's mobile systems are AI-driven: perception, language, decision-making
 - On-device AI is now a core systems problem (efficiency, memory, energy)
 - Mobile LLMs and agents represent the current research frontier
 - Edge-cloud collaboration is central to scalable deployment
 - Robotics and embodied AI inherit mobile computing constraints
- → Studying mobile computing today naturally means studying **mobile AI and on-device AI**

Topics We Will Cover

- On-Device AI
- Model Compression
- Adaptation & Personalization
- Cloud-Edge Collaborative AI
- Mobile LLMs, VLMs, VLAs, Agents
- **Mobile HCI & Applications**
 - Today's topic.
 - We will start from a high-level perspective of the field and then move toward more technical aspects in the later lectures

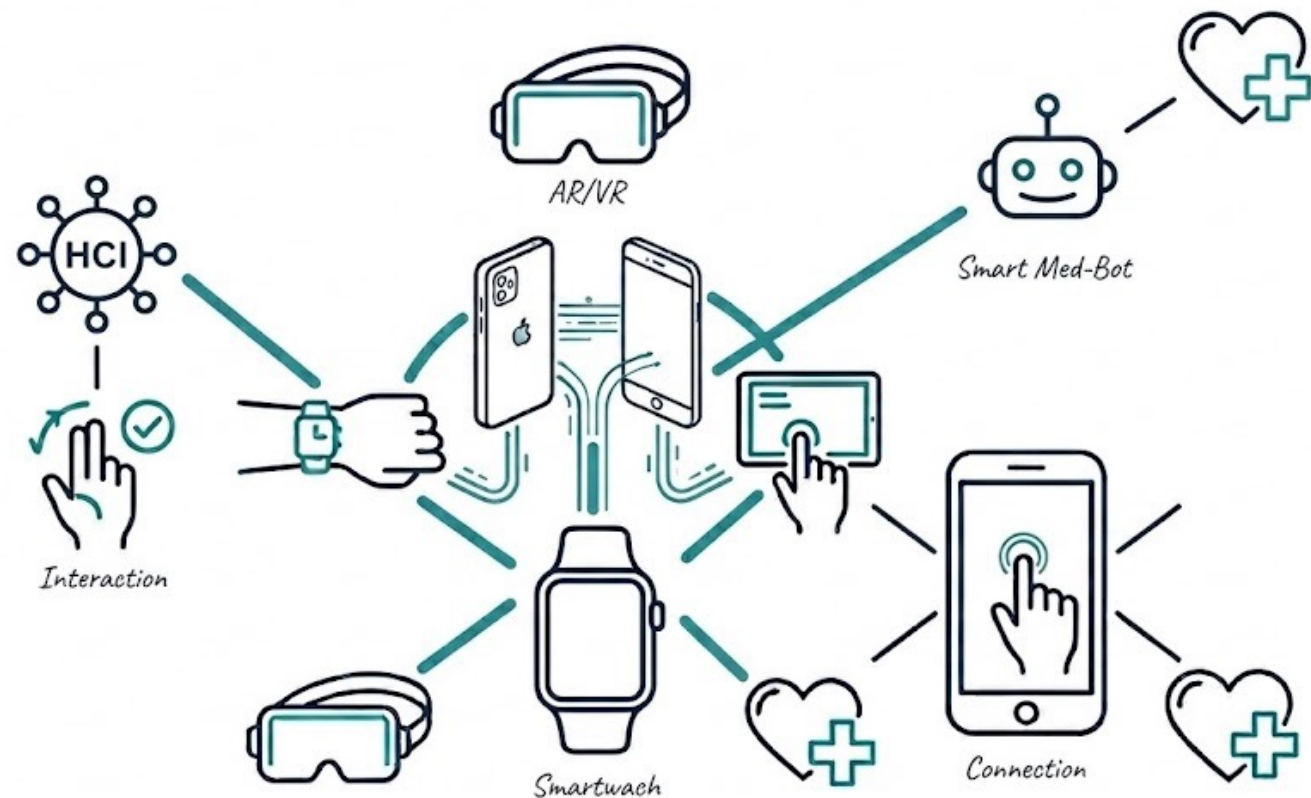
Mobile HCI & Applications

Mobile HCI & Applications

- **Mobile Human-Computer Interaction (HCI) & Applications** examines how people interact with **mobile** and **embedded computing systems** (e.g., smartphones, wearables, robots) and how these systems enable **innovative, context-aware applications** that improve user experience in real-world environments.

- Example Topics:

- Sensing
- Wearables
- Interaction
- Recognition
- VR/AR
- Healthcare



Research Videos

- I am going to show several **research demo videos** related to Mobile HCI and Applications
- Since Mobile HCI and Applications cover **a very broad range of topics** and research directions, it is often easier to understand the field through concrete examples rather than definitions alone
- These videos are based on research papers published in top-tier venues
- I will not cover details of each technology, so please take a look at the paper if you are interested
- By watching them, you can get **an overall sense of what Mobile HCI and Applications** look like in practice, and they may also **inspire ideas for your team project**

Laput, Gierad, and Chris Harrison. "Sensing fine-grained hand activity with smartwatches." CHI '19



<https://youtu.be/a-lmMjOrbbI>

Gong, Taesik, et al. "Knocker: Vibroacoustic-based object recognition with smartphones." UbiComp '19



https://youtu.be/SyQn1vr_HeQ

Breda, Joseph, et al. "NightLight: Passively Mapping Nighttime Sidewalk Light Data for Improved Pedestrian Routing." CHI '25



<https://youtu.be/daCeWg5aDIM>

Kaur, Sawinder, et al. "CRoP: Context-wise Robust Static Human-Sensing Personalization." *UbiComp '25*



<https://youtu.be/QDjNXcy50-E>

Arakawa, Riku, Jill Fain Lehman, and Mayank Goel. "Prism-q&a: Step-aware voice assistant on a smartwatch enabled by multimodal procedure tracking and large language models." *UbiComp '25*



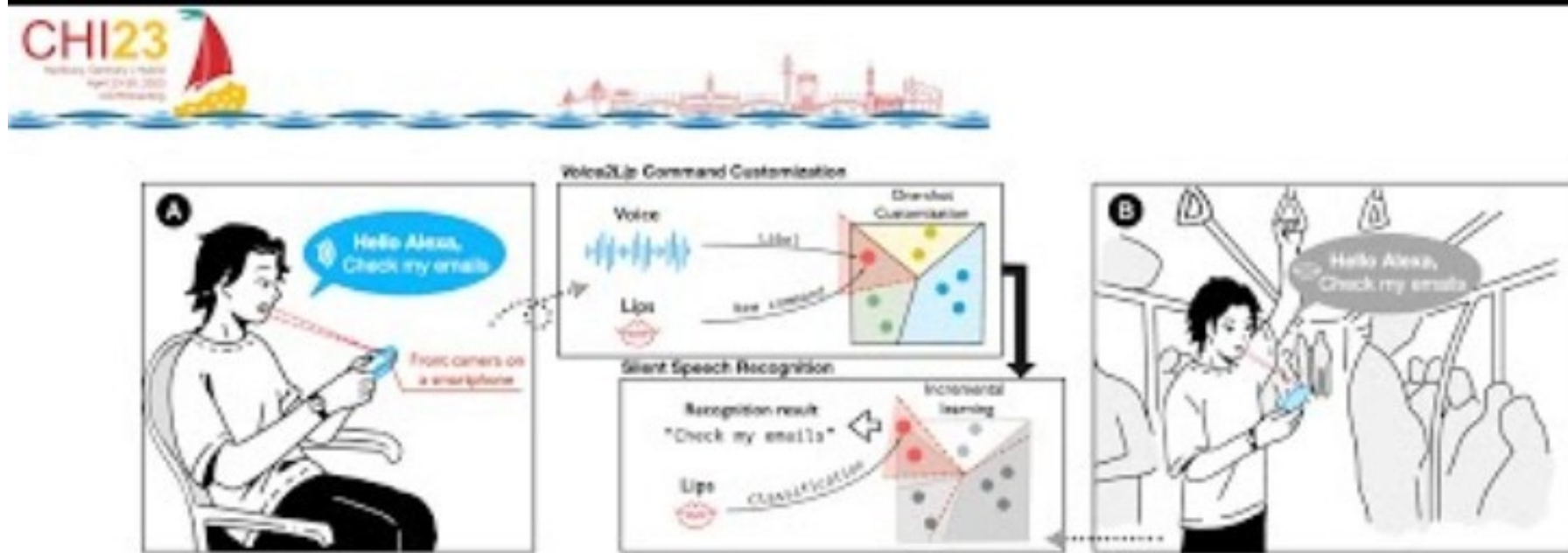
<https://youtu.be/VG4-uZxLFPI>

Kim, Daehwa, Vimal Mollyn, and Chris Harrison. "WorldPoint: Finger Pointing as a Rapid and Natural Trigger for In-the-Wild Mobile Interactions." ISS '23



<https://youtu.be/W42EgodN5Hg>

Su, Zixiong, Shitao Fang, and Jun Rekimoto. "Liplearner: Customizable silent speech interactions on mobile devices." CHI '23



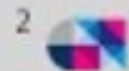
LipLearner

Customizable Silent Speech Interactions on Mobile Devices

Zixiong Su¹, Shitao Fang¹, Jun Rekimoto^{1,2}



THE UNIVERSITY OF TOKYO



Sony CSL

Mollyn, Vimal, and Chris Harrison. "EgoTouch: On-body touch input using AR/VR headset cameras." *UIST '24*

EgoTouch

On-Body Touch Input Using
AR/VR Headset Cameras



<https://youtu.be/7Jwglm1f1ss>

Waghmare, Anandghan, et al. "Glucoscreen: A smartphone-based readerless glucose test strip for prediabetes screening." UbiComp '23



<https://youtu.be/3nJRdWHI6lg>

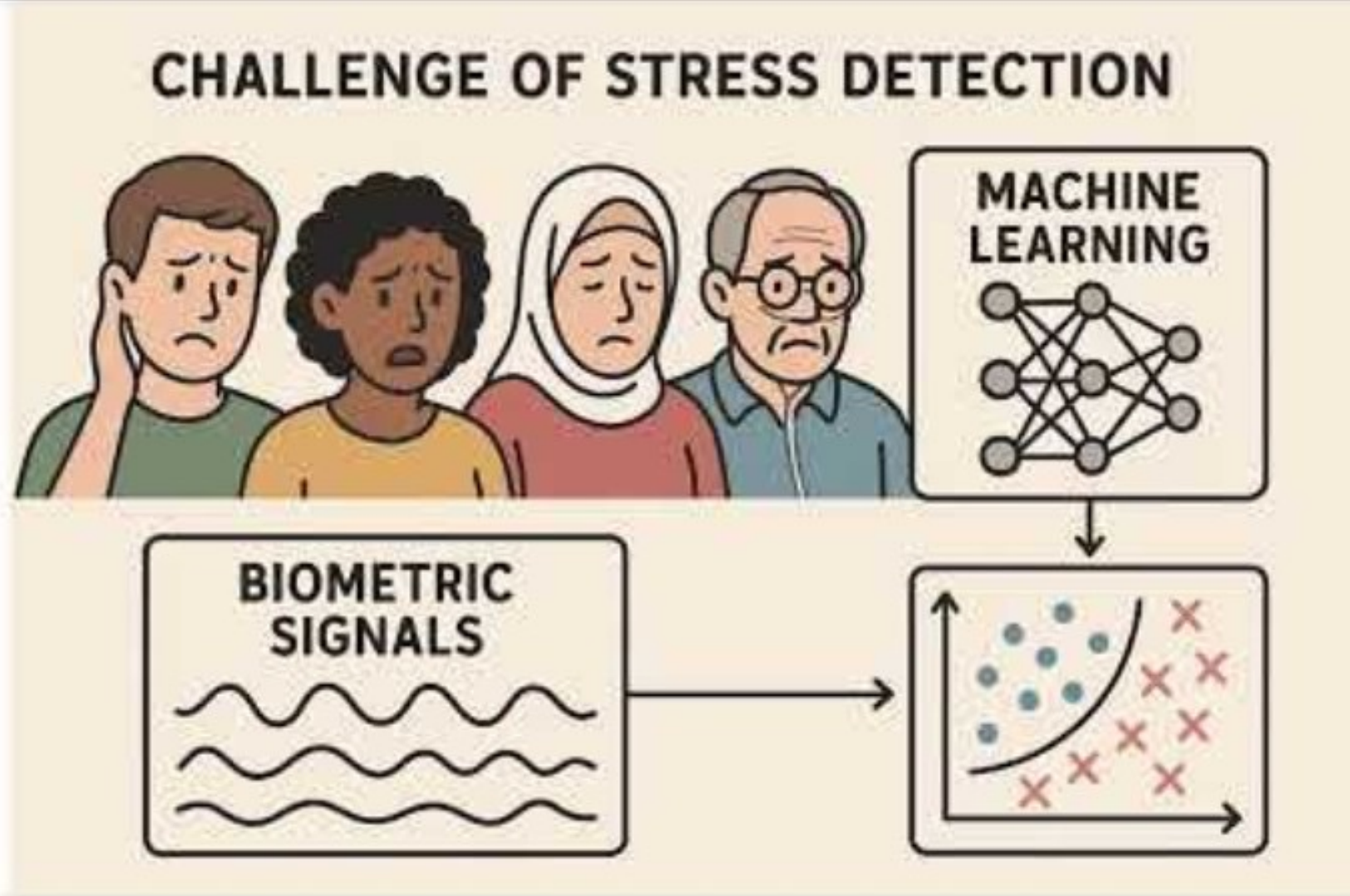
Xue, Qiuyue Shirley, et al. "Luckychirp: Opportunistic respiration sensing using cascaded sonar on commodity devices." *PerCom '22*



In contrast, LuckyChirp operates in the background and automatically detects the user's respiration

<https://youtu.be/1jCMbreMSkk>

Xiao, Yi, et al. "Human Heterogeneity Invariant Stress Sensing." *UbiComp '25*



<https://youtu.be/sEeZSKSITYY>

Zheng, Jian, Subin Seo, and Eun Kyoung Choe. "From Awareness to Action: Ambient Display and Customizable Attention Signals for Self-regulated Smartphone Usage." *UbiComp '25*

TIMER

Timer for
Internalizing
Motivation &
Encouraging
Reflection

That's why we designed TIMER

<https://youtu.be/P6dx7mlgSms>

Islam, Towhidul, et al. "H2opulse: Smartphone-assisted vein evaluation for early recognition of dehydration." *UbiComp '25*

We aim to to develop a dehydration detection system that is -



Dsouza, Vivian, et al. "Densor: An intraoral battery-free sensing platform." *UbiComp '24*



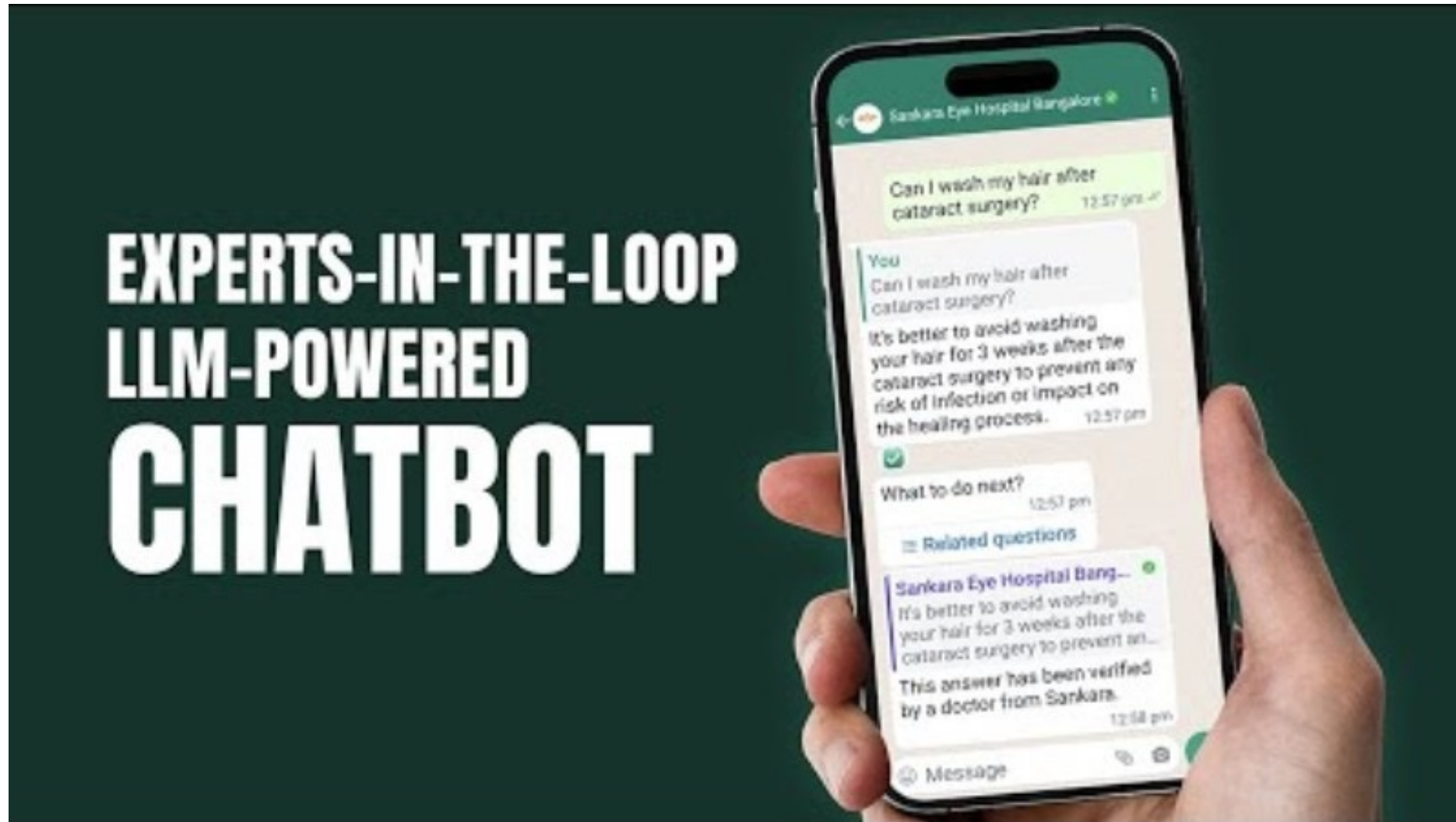
<https://youtu.be/XBBFT1IY6Y>

Yu, Xiaofan, et al. "Sensorchat: Answering qualitative and quantitative questions during long-term multimodal sensor interactions." *UbiComp '25*



https://www.youtube.com/watch?v=wqll0_W2k7s

Ramjee, Pragnya, et al. "CataractBot: an LLM-powered expert-in-the-loop chatbot for cataract patients." *UbiComp '25*



<https://youtu.be/8HeJ69hZ6Gg>

How to Come Up with a Project Idea?

- The topic **must be relevant to mobile computing**
- Consider the areas we cover in this course:
 - On-Device AI
 - Model Compression
 - Adaptation & Personalization
 - Cloud-Edge Collaborative AI
 - Mobile LLMs, VLMs, VLAs, Agents
 - Mobile HCI & Applications

Possible Topics

- Create an **interesting mobile AI application** (similar to the research videos you watched today)
- **Improve the functionality** of existing mobile apps/products (e.g., adding a new AI-powered feature)
- **Improve the performance** of a prior research work (e.g., accuracy, latency, memory usage)
- **Overcome the limitations** of an existing research work (e.g., applying it to a new environment)